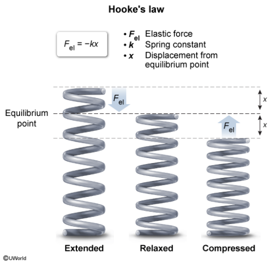
A spring is extended 15 cm from its equilibrium point. If the spring constant k is 75 N/m, the magnitude and direction of the elastic force F_{el} are described by which of the following?

- A. 1.1×10^1 N; oriented away from the equilibrium point [16%]
- B. 1.1×10^1 N; oriented toward the equilibrium point [60%]
- C. 1.1×10^3 N; oriented away from the equilibrium point [10%]
- D. 1.1×10^3 N; oriented toward the equilibrium point [12%]

Correct
61% answered correctly

Explanation:



An **elastic force (F_{el})** is generated when displacing a spring or other elastic object from its **equilibrium point**, which is the natural, nondisplaced position of an elastic object. According to **Hooke's law**, F_{el} is a function of the **object displacement (x)** and the elastic constant (k). This is expressed mathematically as:

$$F_{el} = -kx$$

The value of k quantifies the relative elasticity of an object, which is attributable to its shape and the molecular structure of the material from which it is created. Although no material is perfectly elastic, Hooke's law is still useful in modeling the elastic forces generated by springs and other elastic objects.

When applying Hooke's law, F_{el} is typically expressed as a negative (−) quantity because F_{el} acts counter to the direction of displacement. For example, extending a spring (ie, lengthening it) requires a displacing force (F) to perform some mechanical work on the spring over a positive (+) distance. Because the spring will rapidly return to its equilibrium position and undergo harmonic oscillations if F is withdrawn, F_{el} opposes F and is therefore expressed as a negative quantity.

In this question, a spring with an elastic constant (spring constant) of 75 N/m is extended 15 cm from its equilibrium point. When converted to meters, the displacement of the object due to stretching is:

$$x = 15\ \cancel{\text{cm}} \cdot \frac{1\ \text{m}}{100\ \cancel{\text{cm}}}$$

Using Hooke's law to calculate F_{el} gives:

$$F_{el} = -\left(75\ \frac{\text{N}}{\cancel{\text{m}}}\right) \cdot (0.15\ \cancel{\text{m}})$$

$$F_{el} = -11\ \text{N} =$$

Consequently, the magnitude of F_{el} is:

$$|F_{el}| = 1.1 \times 10^1\ \text{N}$$

Furthermore, F_{el} points *toward* the equilibrium point because F_{el} acts to oppose the extension and restore the stretched spring back to its shorter natural position.

Therefore, extending the spring 15 cm from its equilibrium point will generate an elastic force that has a magnitude of 1.1×10^1 N and is oriented toward the equilibrium point **(Choice B)**.

Educational objective:
Hooke's law may be used to calculate the elastic forces generated by the compression and extension of springs and other elastic objects. The magnitude of the elastic force varies with displacement, but elastic forces are always oriented toward the equilibrium point.

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